

Naive Bayes & LDA — Engineering Report

Project 008 · Dual-Dataset Classification with Gaussian Naive Bayes & Linear Discriminant Analysis

Project Overview

This project explores **Naive Bayes classification** and **Linear Discriminant Analysis (LDA)** for dimensionality reduction across two semiconductor/IT manufacturing datasets. GaussianNB is trained on each dataset independently with StandardScaler preprocessing, and LDA vs PCA comparisons demonstrate supervised vs unsupervised dimensionality reduction.

Datasets

Dataset	Rows	Features	Target	Classes
Jira Bug Routing	8,000	6 numeric	category	6 (Hardware, Firmware, Software, Performance, Security, Docs)
Silicon Defect Classification	6,000	5 numeric	defect_type	7 (6 defect types + Clean Die)

Methodology

- **Preprocessing:** StandardScaler normalization of all features
- **Model:** GaussianNB — assumes features follow a normal distribution per class
- **Dimensionality Reduction:** LDA (supervised) vs PCA (unsupervised) comparison
- **Evaluation:** Accuracy, precision, recall, F1, confusion matrices
- **Serialization:** joblib pipeline export (gnb_jira.pkl, gnb_silicon.pkl)

Results

Jira Accuracy: 88.6%

Silicon Accuracy: 99.9%

Jira Bug Routing — Per-Class Performance

Class	Precision	Recall	F1
Documentation	0.98	0.96	0.97
Firmware	0.73	0.74	0.74
Hardware	0.94	0.94	0.94

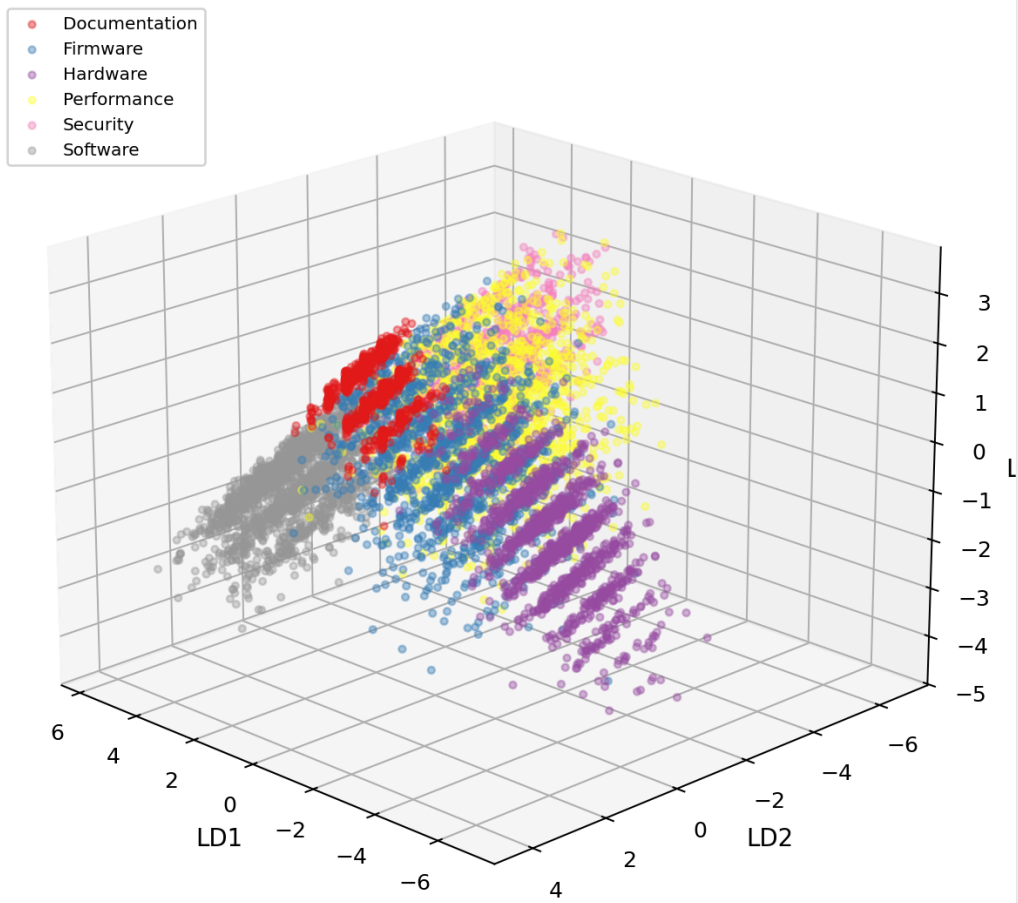
Performance	0.79	0.72	0.76
Security	0.88	0.93	0.90
Software	0.94	0.96	0.95

Silicon Defect Classification — Per-Class Performance

Class	Precision	Recall	F1
Clean Die	1.00	1.00	1.00
Gate Oxide Trap	0.99	1.00	1.00
Metal Short	1.00	1.00	1.00
Oxide Pinhole	1.00	0.99	1.00
Particle Defect	1.00	1.00	1.00
Via Open	1.00	1.00	1.00

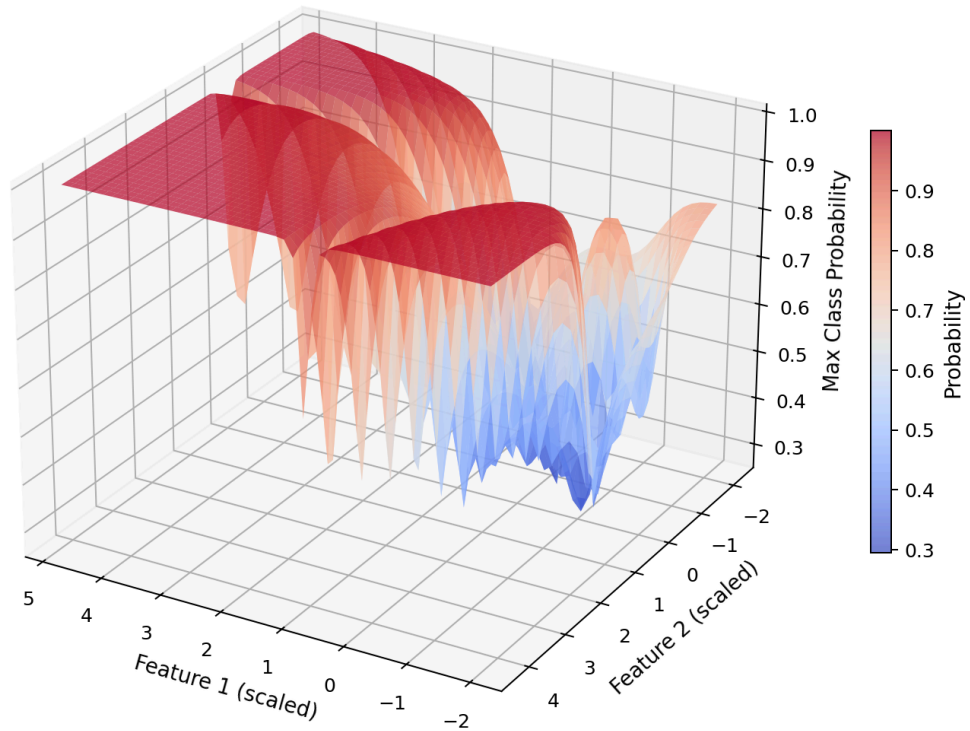
Project 1 — Jira Bug Routing Visualizations

LDA 3D Projection — JIRA Bug Categories



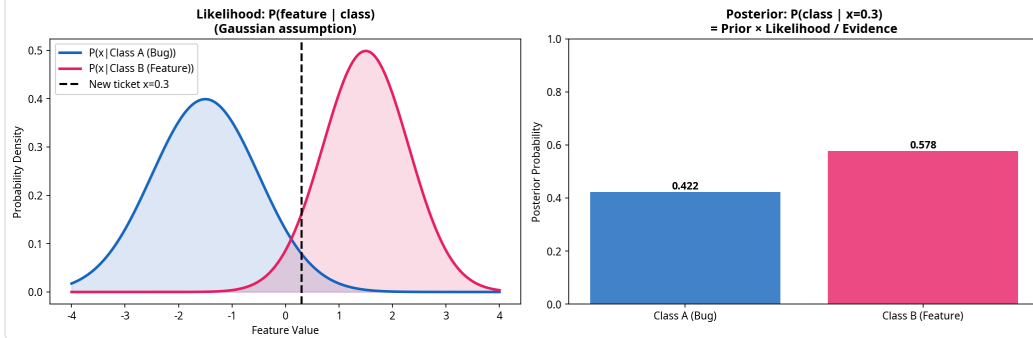
Proj1 Jira 3D Lda

Naive Bayes Posterior Probability Surface JIRA Bug Routing



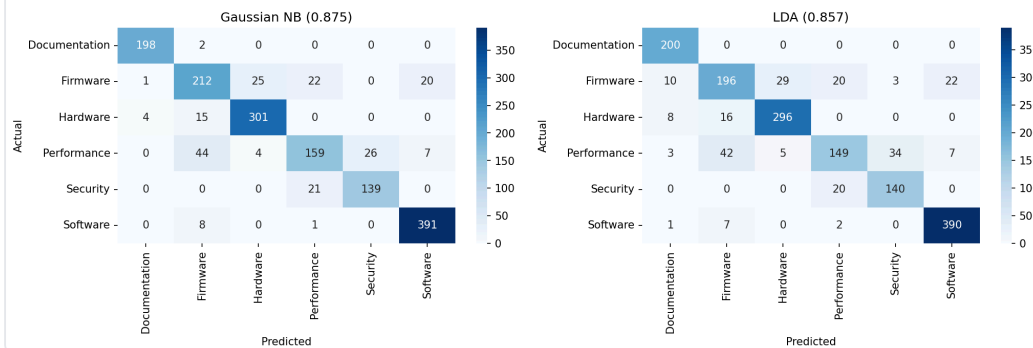
Proj1 Jira 3D Nb Probability

Bayes' Theorem in Action: How Naive Bayes Classifies a New Ticket

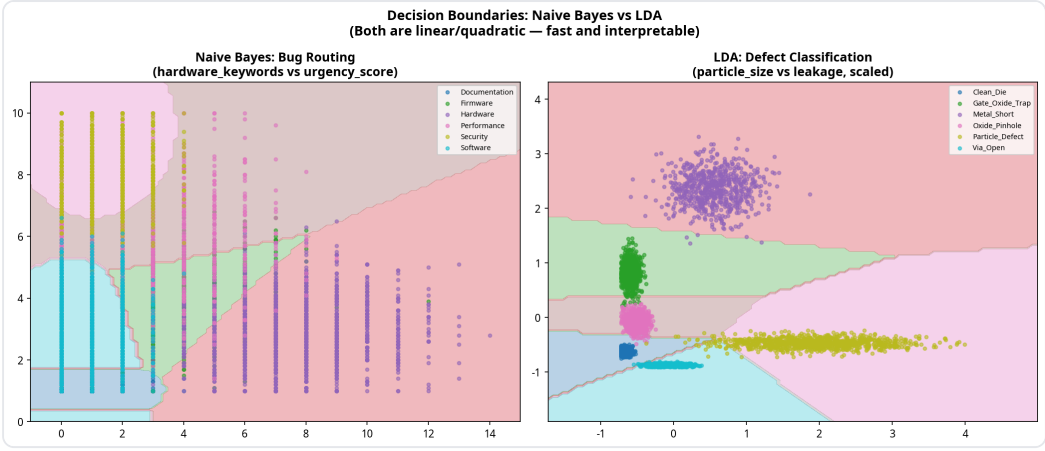


Proj1 Jira Bayes Theorem

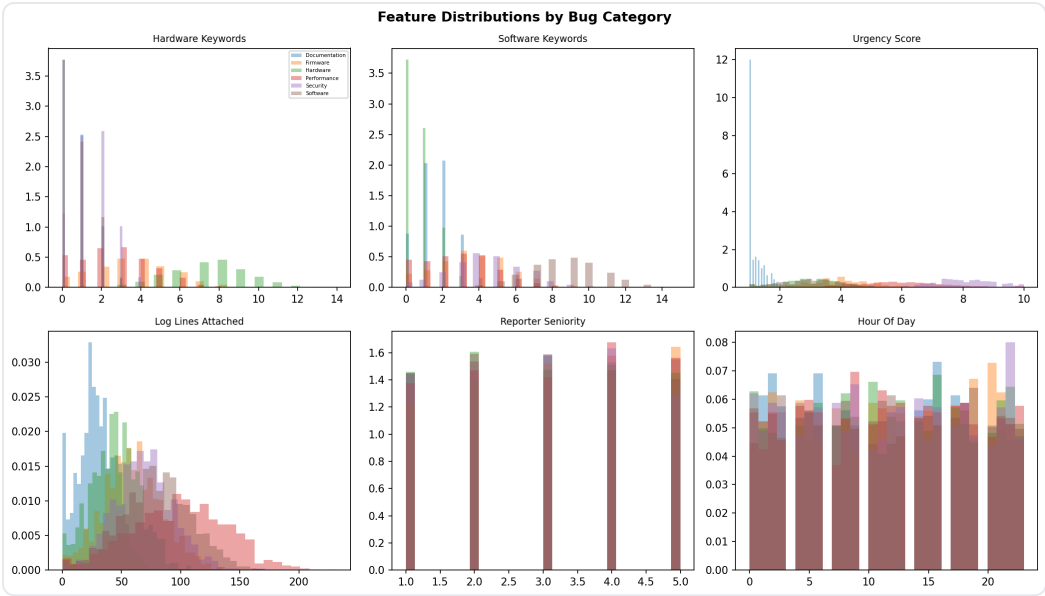
Confusion Matrices



Proj1 Jira Confusion

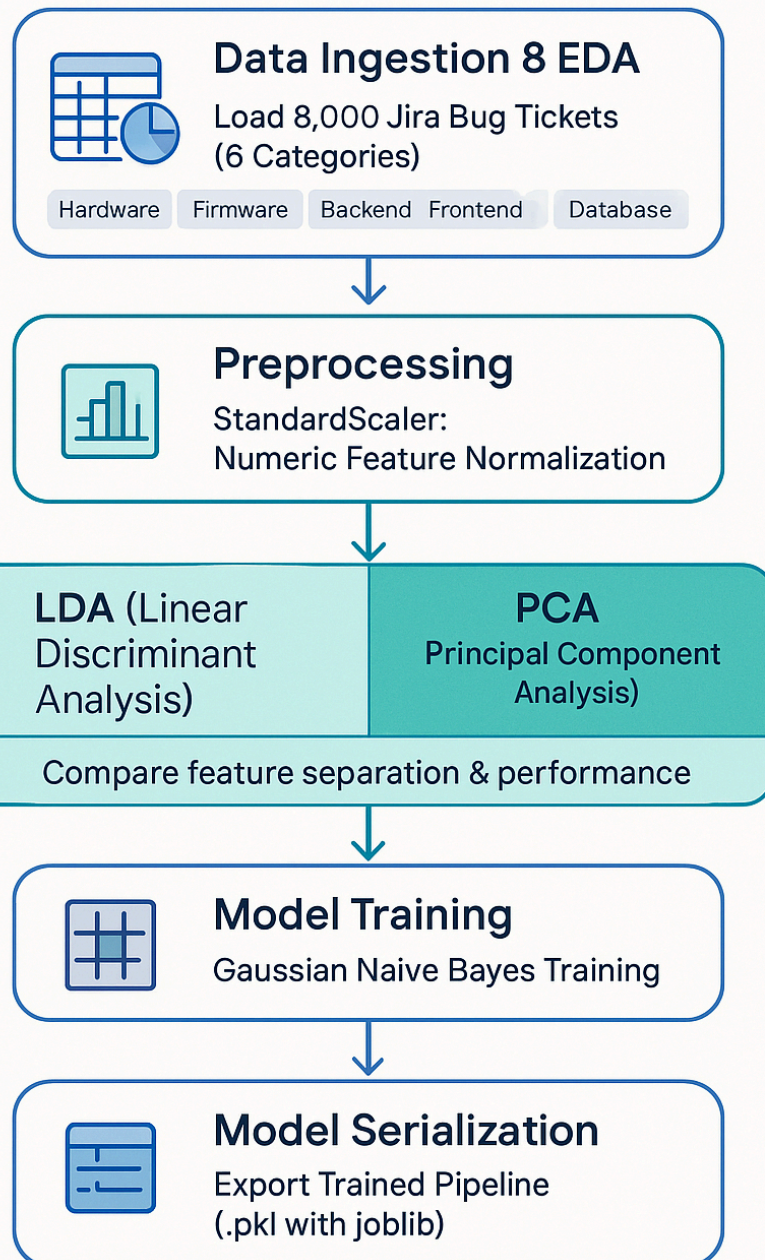


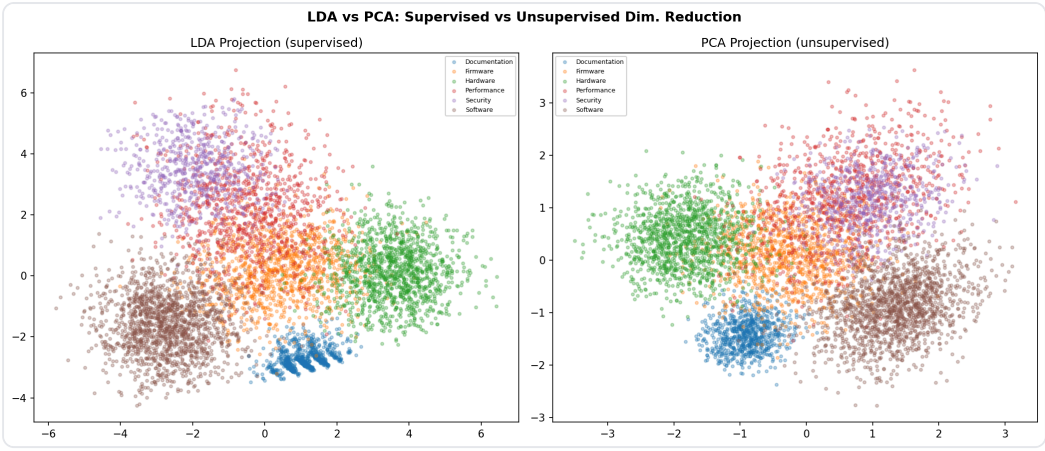
Proj1 Jira Decision Boundaries



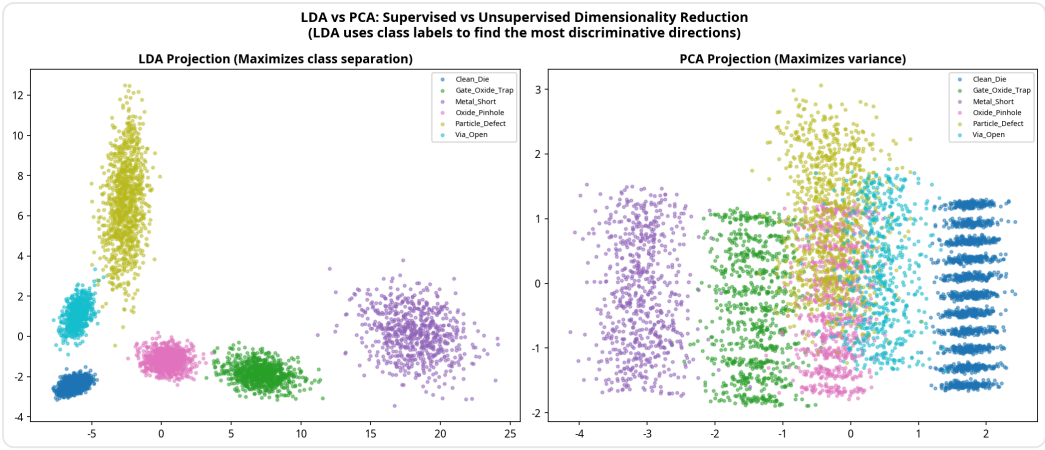
Proj1 Jira Eda

Jira Bug Routing – Naive Bayes & LDA Pipeline

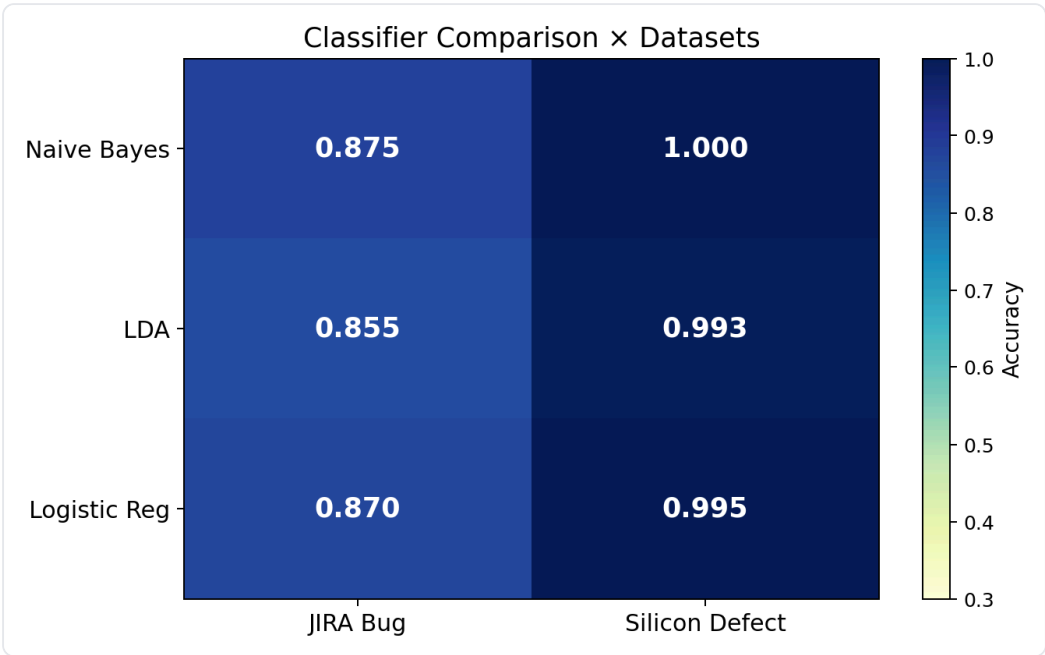




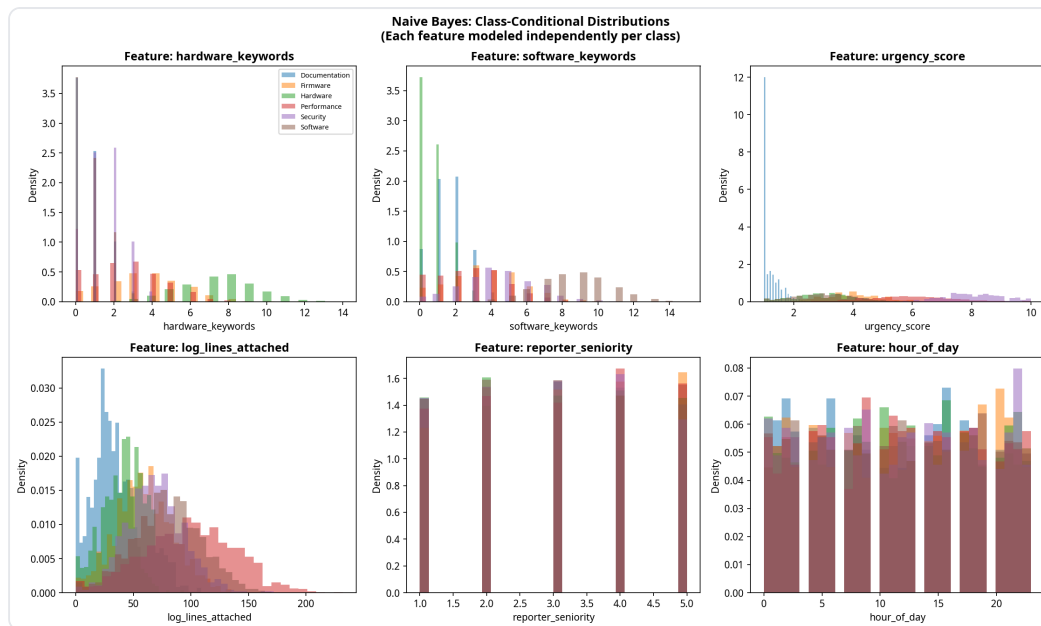
Proj1 Jira Lda Vs Pca



Proj1 Jira Lda Vs Pca Comparison



Proj1 Jira Model Heatmap

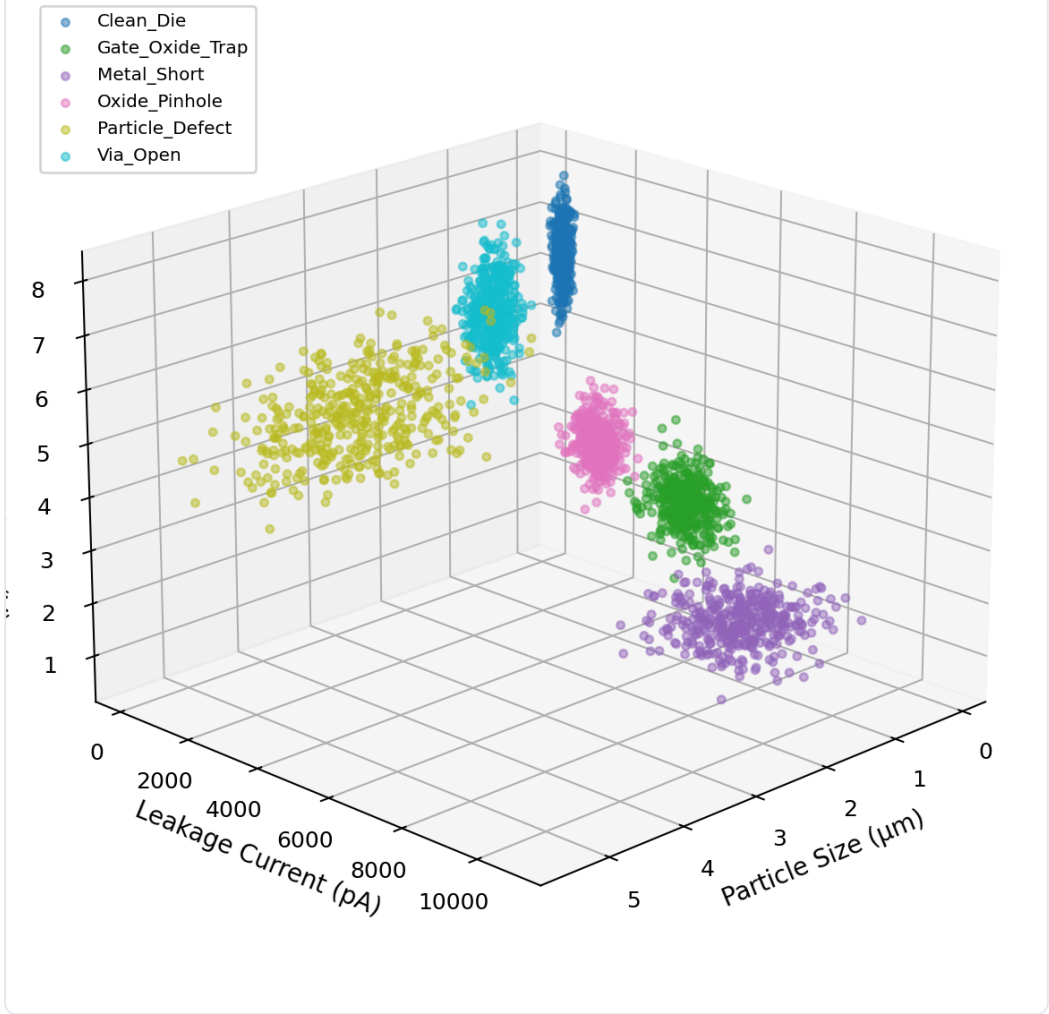


Proj1 Jira Nb Distributions

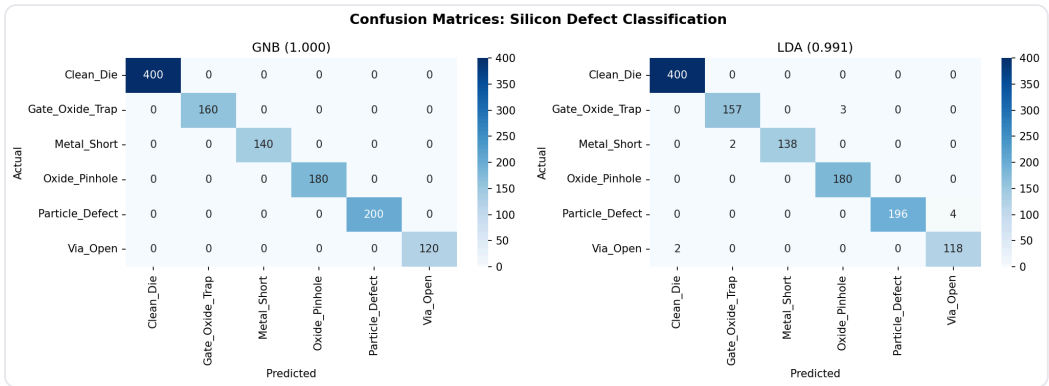
Project 2 — Silicon Defect Classification Visualizations

3D Silicon Defect Feature Space

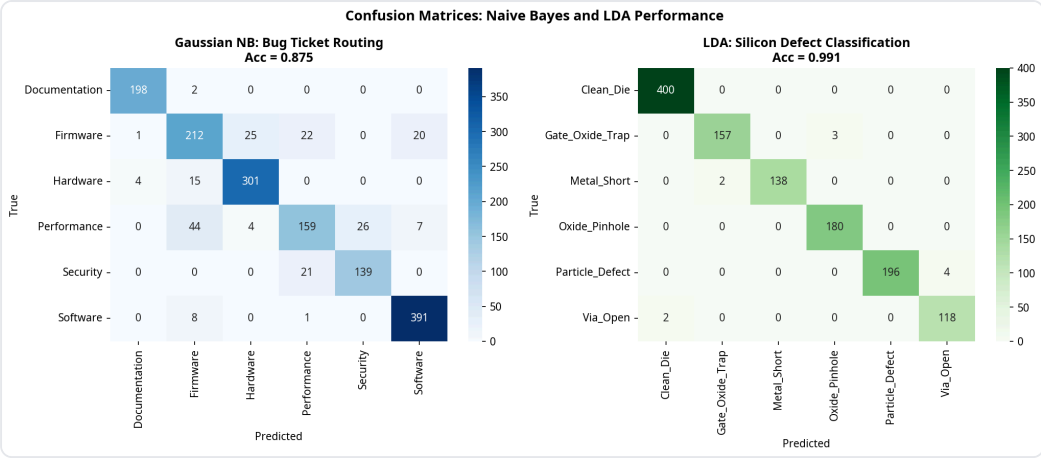
Color = Defect Type



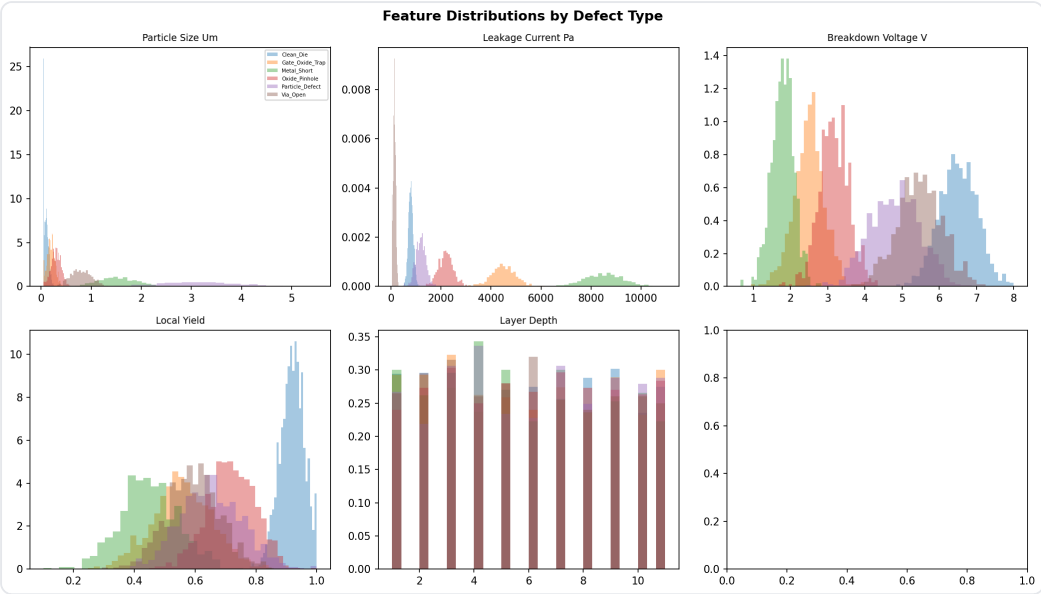
Proj2 Silicon 3D Defects



Proj2 Silicon Confusion



Proj2 Silicon Confusion Matrices



Proj2 Silicon Eda

Silicon Defect Classification – Naive Bayes & LDA Pipeline



Data Ingestion & EDA

6000 rows, 5 features – wafer
measurements & defect labels

Class balance, feature stats, defect type distribution



Preprocessing

StandardScaler – Feature normalization

Ensures GaussianNB assumptions,

mean=0, std=1



Dimensionality Reduction

Linear Discriminant Analysis (LDA)

Maximizes class
separation – uses label

Principal Component Analysis (PCA)

Captures maximum
variance – unsupervised

Comparison for visualization & insight



Model Training

Gaussian Naive Bayes – Multiclassifier

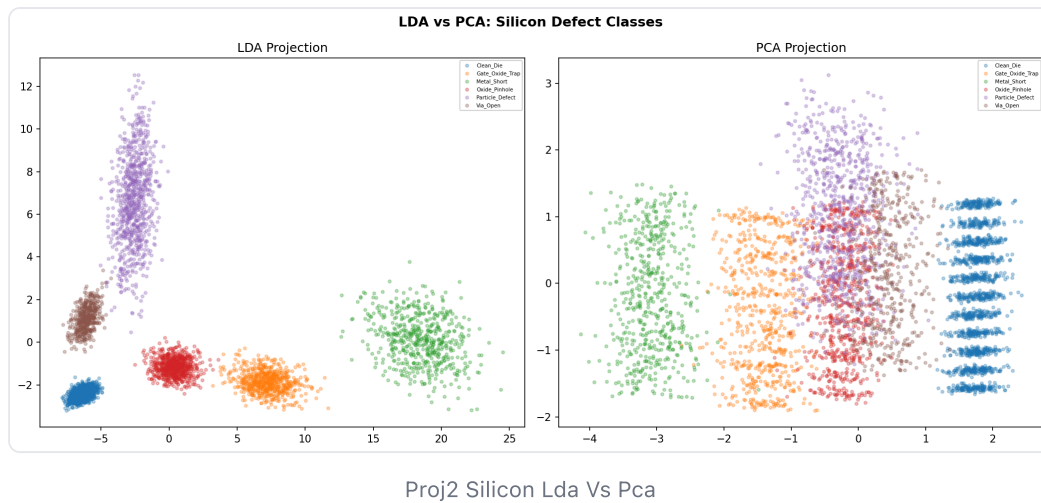
Trained on preprocessed and reduced data



Model Serialization

Export model – joblib (.pkl)

Ready for deployment & inference



Architecture

- **Training:** `src/train.py` — DATASETS dict pattern, trains both models
- **Inference:** `src/predict.py` — loads any model, demos both datasets
- **API:** `src/api.py` — FastAPI endpoint serving GaussianNB (Jira default)
- **Testing:** `tests/test_model.py` — 4 tests (existence + prediction × 2)